

JKBOSE 2026

CLASS 11TH

PLANT KINGDOM

LECTURE-06

BOTANY

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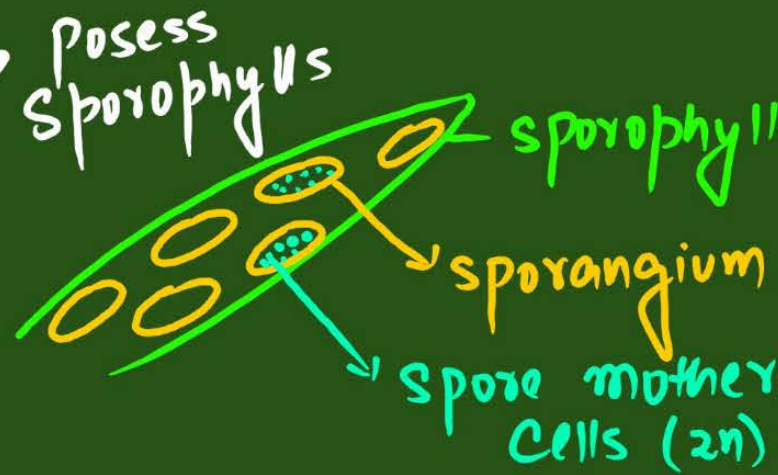


Topics To Covered

- 1 Life Cycle of Homosporous Pteridophytes ✓
- 2 Life Cycle of Heterosporous Pteridophytes ✓
- 3 General Characters of Gymnosperms ✓ } *Next lecture....*

LIFE CYCLE OF HOMOSPOROUS PTERIDOPHYTES

Main plant Body
Sporophyte ($2n$)
Ex. Ferns



meiosis

Spores (n)
(same kind)

Germinate

- Avascular
- possess rhizoids
- Requires shade for growth

Gametophyte (n)
(prothallus)

mitosis

Fertilization
(zooidogamy)

Archegonium

Egg cell (n)

Prothallus

Anthrezoid (n)
(♂ gamete)
Anthredium

Gametophyte (n)

mitosis

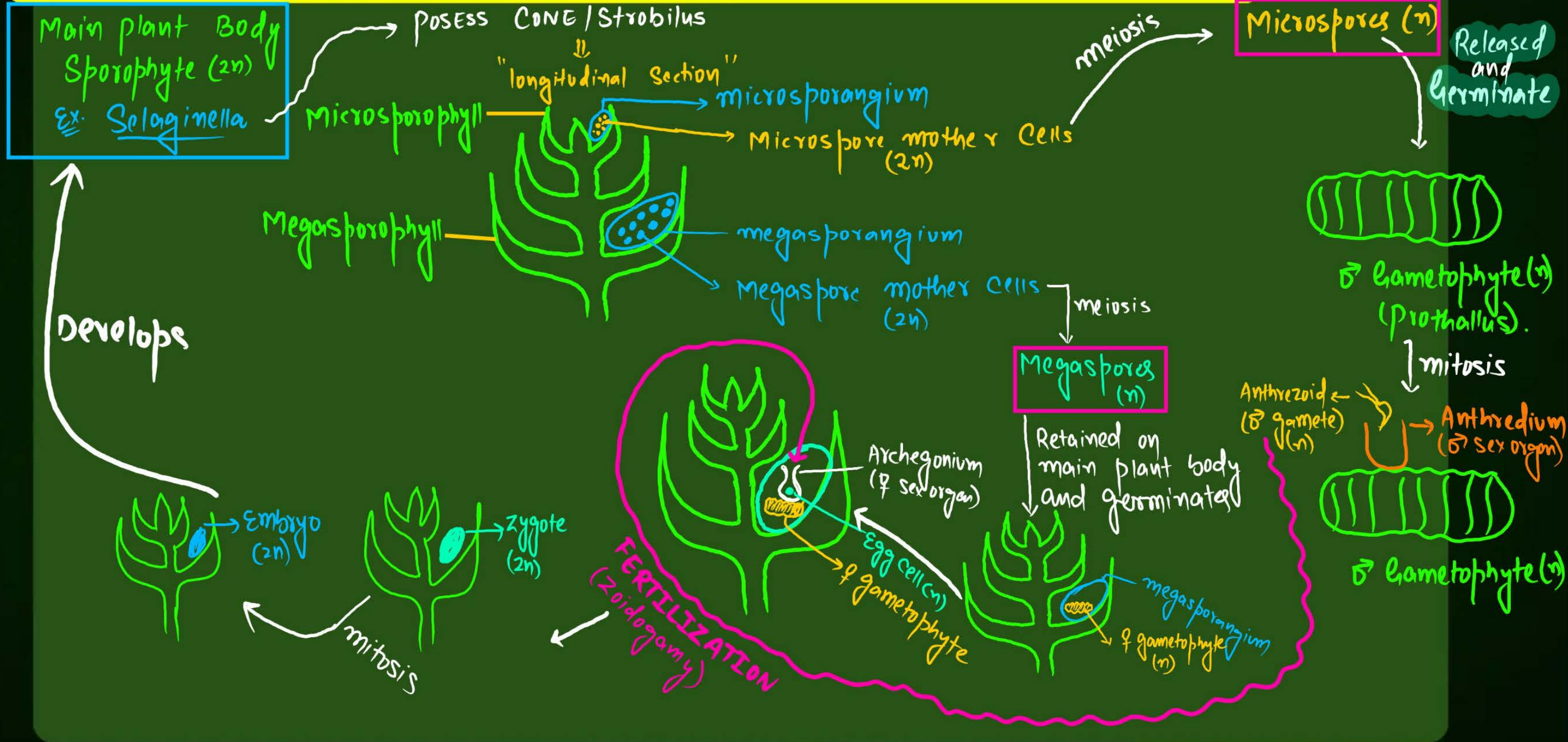
Embryo ($2n$)

Zygote ($2n$)

Develop.

Note:- In homosporous pteridophytes — Gametophyte i.e., prothallus is
monoecious

LIFE CYCLE OF HETEROSPOROUS PTERIDOPHYTES



Note :-

1. $9n$ heterosporous pteridophytes — Gametophyte is Dioecious.

2. $9n$ " " — ♀ gametophyte is retained on main plant body (i.e., Sporophyte).



This is an important Step in Evolution

⊗. Heterosporous pteridophytes — are the precursors of seed habit.
 ↓
 "Seed formation"

Explanation:- Following Steps are required for seed formation:

- | | | |
|--|---|---|
| Fulfilled in
heterosporous pterido-
phytes | { | ①. Plant should be heterosporous. |
| | | ②. ♀ gametophyte should be retained on main plant body. |
| Not - fulfilled | { | ③. Presence of protective covering (Kld integuments) should be int around megasporangium. |
| | | ④. Zygote should undergo — resting stage. |

Note:-

1. Gametophyte (prothallus) in pteridophytes requires shade for growth.
2. Pteridophytes requires H_2O for fertilisation.

Reasons that why pteridophytes are limited in distribution.

CLASSIFICATION OF PTERIDOPHYTES

04 Orders of pteridophytes

1. Psilopsida e.g, *Psilotum* ✓
2. Lycopsida e.g, *Selaginella*, *Lycopodium* ✓
3. Sphenopsida e.g, *Equisetum* (horse tail)
4. Pteropsida e.g, *Dryopteris*, *Pteris*, *Adiantum*



Q. Which among the following are heterosporous plants?

A. Selaginella, Salvinia — (heterosporous pteridophytes)

B. Gymnosperms. } seed + nt i.e., they are heterosporous.

C. Angiosperms

✓ D. All of the Above.

QUESTIONS

Seed habit first established in:

Seed formation

- ☐ A Pteridophytes
- ☒ B Gymnosperms ✓
- ☐ C Angiosperms
- ☐ D None of the above

QUESTIONS

Funaria and Salvinia are
Moss Homosporous Het. Pteridophyte

- ☐ A Homosporous
- ☒ B Homosporous and Heterosporous respectively
- ☐ C Heterosporous and Homosporous respectively
- ☐ D Heterosporous

QUESTIONS

Assertion (A): Pteridophytes are successful land plants. *TRUE.*

Reason (R): Pteridophytes possess vascular tissue and roots. *TRUE*

- ☒ **A** Both (A) and (R) are true and the (R) is the correct explanation of the (A)
- ☐ **B** Both (A) and (R) are true but the (R) is not the correct explanation of the (A)
- ☐ **C** (A) is true but (R) is false
- ☐ **D** Both (A) and (R) are false

QUESTIONS

Female gametophyte (n) retained on sporophyte ($2n$) first of all in plant kingdom seen in:

- ☐ A Algae
- ☐ B Bryophyte
- ☒ C Pteridophyte
- ☐ D Gymnosperm

QUESTIONS

What is the reason of limited distribution of pteridophyte?

- ☐ A Habitat
- ☐ B Water for fertilization
- ☐ C Absence of vascular tissue
- ☒ D Habitat and water for fertilization

GYMNOSPERMS

↓
"Next lecture"

Thank
You